This problem set is not going to be collected, although solutions will be posted before the quiz. In addition to solving them, I recommend writing up solutions as if you were going to submit them. Unlike a problem set collected for credit, though, you are free to discuss with classmates and course staff and get help.

I recommend you first attempt the problems on your own, then discuss with classmates. After that, you may want to consult course staff at help hours or on Ed Discussion. After solutions are posted, consult the solutions and compare to what you wrote, and then discuss the differences with your study group.

Quiz 1 will have two free response questions; one will be to use divide and conquer to solve a problem, similar to group one on this problem set. The other will be to use a graph to solve a problem, similar to group two on this problem set.

1 Divide and Conquer

1. In the lecture, we discussed the "famous problem." We saw an algorithm that uses exactly $3n - 3$ questions to identify a celebrity.

   (a) Is it true or false that any correct algorithm to identify a celebrity in a group of size n must ask $\Omega(n)$ questions? Justify your response in a few sentences. You may assume I am only asking about deterministic algorithms if you would prefer.

   (b) Give an algorithm to identify a celebrity with at most $3n - 3 - \log n$ questions asked, or correctly determines that no celebrity exists in the input group given. Note that this is strictly fewer than the algorithm given in class, but is not asymptotically better. You may assume that n is a power of two for purposes of this problem. Explain how your algorithm uses the desired number of questions.

2. Suppose I have an array A of n integer elements. Given a value x, I wish to find out if there are indices i and j such that $A[i] + A[j] == x$.

   For example, my value x might be 57 and my array might be:

-5	30	2	15	96	42

   In which case, I would answer true.

   You are allowed to make comparisons between elements, but cannot otherwise use their exact values. For example, do not use anything like hashing or radix sort.

   (a) The following algorithm correctly solves this problem. What is the running time? Use Θ notation.

   ```
   for i ← 1 to n do
     for j ← i + 1 to n do
       if A[i] + A[j] == x then
         return true
   return false
   ```

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(b) Give an algorithm to solve this problem in time $\mathcal{O}(n \log n)$. You may allocate $\mathcal{O}(1)$ additional memory in doing so. It is okay if your running time is *in expectation* instead of guaranteed, but you must state such in your answer if this is the case.

(c) Suppose that A is already sorted. Give an algorithm to solve this in time $\mathcal{O}(n)$.

3. Consider the problem of finding a *local minimum* in an array. The value $A[i]$ is a local minimum if - and only if - $A[i-1] \geq A[i] \leq A[i+1]$. Give an efficient **divide and conquer** algorithm to find a local minimum in an array A in which $A[1] = A[n] = \infty$. State the runtime of your algorithm, both as a recurrence relation and in asymptotic notation and briefly (1-2 sentences is sufficient) explain the correctness of your algorithm.

   For example, if your input array A is:

∞	10	23	17	20	1	2	3	4	-3	5	∞

   then any of the grayed cells could be returned as a valid answer (you need only to find *one* local minimum, not all of them).

   A solution with running time $\Omega(n)$ will receive no credit for this problem.

4. The Sorting Hats problem is as follows. I have a collection of n hats and n mannequins (statues that resemble a human and are used to model clothing and accessories). My goal is to put one hat on each mannequin head. Each hat is a distinct size, and no two heads are the same size. For each hat, there is exactly one head that the hat fits perfectly. I cannot directly compare two hats or two heads, nor can I inspect a single hat or a single head to determine its size.

   I can, however, try to place a hat on a head. If I do so, I find out one of the following pieces of information:

   - The hat fits the head perfectly
   - The hat is too small for the head
   - The hat is too big for the head

   We can then remove the hat from the head if we so choose.

   The goal of the Sorting Hat problem is to match each head with the hat that fits it perfectly while placing as few hats on heads as possible. Give an approach to do so. If there are n hats and n heads, express the number of times your approach will place a hat on a head in $\mathcal{O}$-notation and justify why that many such operations will happen. Clearly indicate if you are measuring *worst-case* or *expected case* (either is okay, but be sure to clearly indicate it).

   Good credit is available for any reasonably-efficient solution, clearly described, with more credit for more efficient solutions.

5. We've all heard of square root, but what about rectangle root? I define the k-rectangle root r of n to be:

$$(r)(r+k) = n$$

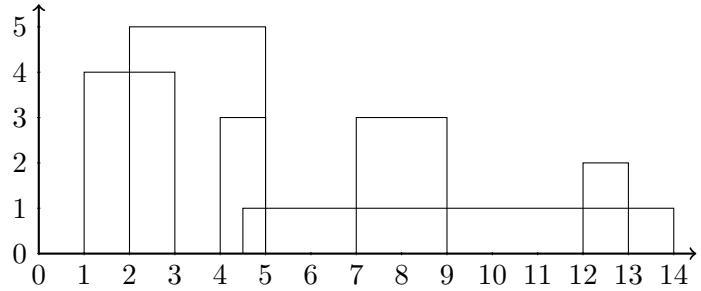
   For example, the 3-rectangle root of 130 is 10 because $10(10+3) = 130$.

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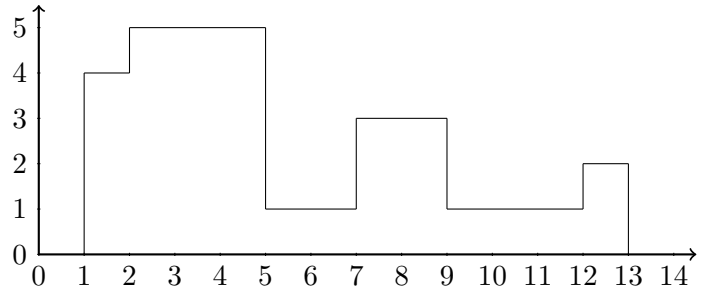
Write or describe an $\mathcal{O}(\log n)$ time algorithm to find the $\lfloor r \rfloor$, the floored k-rectangle root of n, given that $\lfloor r \rfloor, k, n$ are all natural numbers. Do not use a square root operation as part of your solution.

6. (2 points) Suppose you are given a description of rectangular, overlapping buildings and you would like to draw the corresponding skyline with no overlapping lines. A *skyline* is a list of tuples, each consisting of alternating x coordinates and the heights connecting them.

   The **input** is a series of 3-tuples; each tuple describes the left coordinate of the building, the height, and the right coordinate. For example, the input $\{(1, 4, 3), (2, 5, 5), (4, 3, 5), (4.5, 1, 14), (7, 3, 9), (12, 2, 1$ would represent the following buildings:



   You may assume that each building's left-coordinate is strictly before its right-coordinate, that there are no gaps in the buildings, and that each height value is positive. Give an efficient **divide and conquer** algorithm that, given the description of n buildings, forms the corresponding skyline; that is, the overlapping lines are removed. In the case of the above input, the result would be the output (1, 0), (1, 4), (2, 4), (2, 5), (5, 5), (5, 1), (7, 1), (7, 3), (9, 3), (9, 1), (12, 1), (12, 2), (13, 2), (13, 0), which visualized as:



   Give also the running time of your algorithm, both as a recurrence relation and in closed form.

7. You are given a directed graph $G = (V, E)$ with exactly $\frac{n(n-1)}{2}$ edges. For every pair of nodes $u, v \in V$, one of the edges (u, v) and (v, u) are in the graph, and the other is not. You want to find a simple directed path which includes every node in the graph exactly once. It turns out that there is **always** such a path, which you will realize once you find the algorithm. Give an efficient Divide and Conquer algorithm to find such a path, and analyze the running time by setting up and solving a recurrence relation.

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2 Graph Usage

For each of these problems, and for a problem like them that will appear on the first quiz, it is a requirement that your algorithm follow the procedure of either creating a graph or modifying a graph that was provided in input, then calling an existing algorithm that you have seen (such as BFS, DFS, Dijkstra's, Topological Sort), and using the output of that to solve the problem. If your solution is something like "run DFS, but change it as follows," you will earn zero credit for the question.

8. There exists a haunted maze which contains n scare stations, with a designated starting station s and a final station t. To model the haunted maze as a graph, there is a vertex for each scare station and a directed edge from one station to another if it is easy to walk between the two directly (note: because the owners of the haunted house place a restriction on which houses you can visit, the edge between the two scare stations is not bidirectional). Each scare station v has a scare factor of $c(v) > 0$, where the higher $c(v)$ is, the scarier the scare stations are. However, $c(s) = c(t) = 0$. Thus the graph has costs on the vertices rather than the edges. Shindler is a scaredy-cat, and wants to complete the maze by minimizing the total scare factor of the houses; in other words, he wants to find a path P from s to t such that $\sum_{v \in P} c(v)$ is minimum.

   Suppose you already have Dijkstra's Algorithm implemented for directed graphs. Describe how you can use that implementation to solve this problem. For credit, your answer must involve creating a set of edge weights for the same set of vertices and edges such that a call to Dijkstra's algorithm will produce the desired tree. Do not re-create a "Dijkstra-like" algorithm and do not modify the code for Dijkstra's algorithm. Your answer must make a function call to Dijkstra's as a meaningful part of your solution.

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9. Suppose you are given an unweighted directed graph $G = (V, E)$ which may or may not have cycles. The vertices, however, are all designated as azure, navy, or gold. This may or may not be a valid 3-coloring of the graph (and, for this problem, it does not matter if it is). The **secluded factor** of any given vertex v is the **largest** of the following three quantities:

   - The shortest distance from any gold vertex to v
     (that is, among all the gold vertices, whichever has the shortest path from it to v is the quantity for this item)
   - The shortest distance from any azure vertex to v
   - The shortest distance from any navy vertex to v

   If we cannot get to any particular vertex from at least one gold vertex, at least one navy vertex, and at least one azure vertex, then that vertex is *infinitely* secluded.

   Given such a graph, describe how you would find a vertex with the *smallest* secluded factor. Give and briefly justify the running time of your approach. For full credit, your approach should have linear running time. You may make any reasonable assumption about how the graph is represented that you would like; if it affects your algorithm, write the assumption in the answer space.

10. Consider the problem A.J. Brown used to face when driving to work each morning. This problem can be represented by a directed graph $G = (V, E)$ with a distinguished vertex s to represent A.J. Brown's house and another t to represent his destination. A subset of the vertices $C \subset V$ are marked as "Chipotle." A.J. wants to find the shortest path from his house to his destination such that he crosses *at most* one Chipotle. That is, we want the shortest path from s to t that uses at most one vertex from set C. You may assume that at least one such path exists. For full credit, your approach should have running time linear in the size of the original input graph.

    Hint: you are allowed to create a second graph, with approximately twice as many vertices as the input graph. What are the vertices in the new graph? What does being at any particular vertex during a path from s to t mean? How can you create the graph? What can you do once you create the graph?

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